DECISION & IMPACT

Ads vs PSA | 588,101 users | Teal: PSA / amber: ads

LIFT SUPPORTED / Verify economics before scaling

PSA CONVERSION

1.79%

AD CONVERSION

2.55%

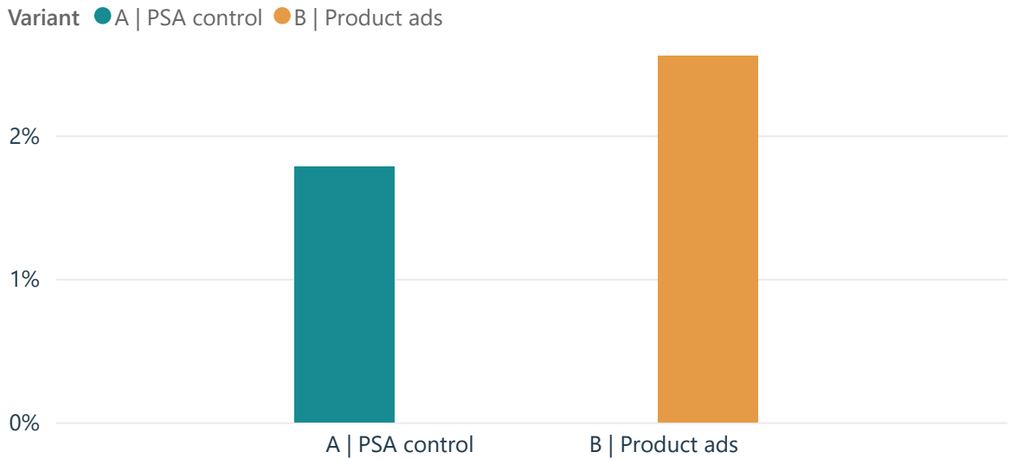
ABSOLUTE LIFT

+0.769 pp

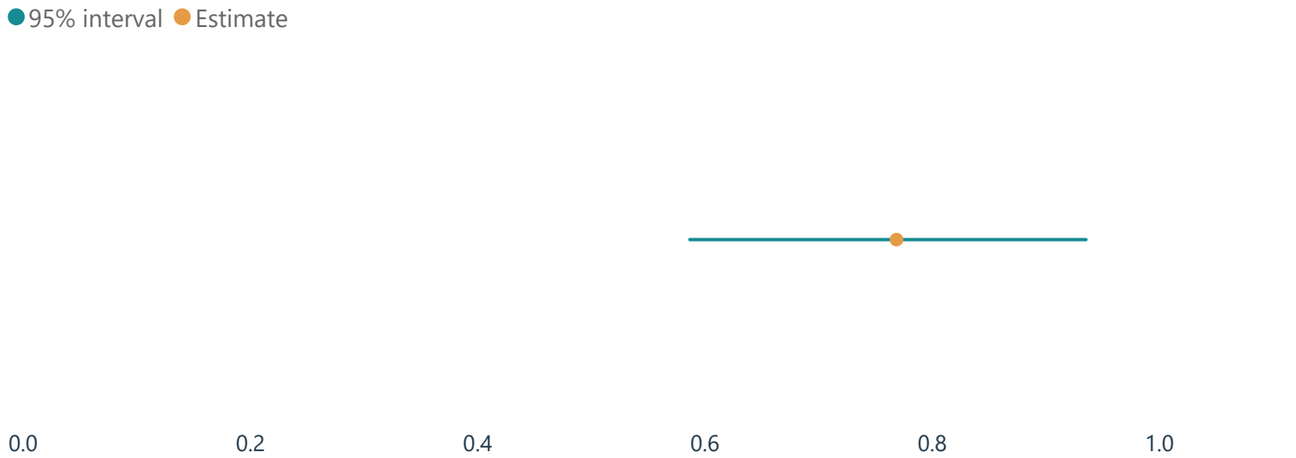
EXTRA CONVERSIONS*

4,343

CONVERSION BY ARM



95% CONFIDENCE INTERVAL | ABSOLUTE LIFT (pp)



CONVERSION EVIDENCE

Positive lift

PROFITABILITY

Not measured

NEXT ACTION

Keep a holdout

*Modeled incrementality; assumes valid assignment. Full-cohort interval: +0.587 to +0.936 pp.

Kaggle: faviovaz/marketing-ab-testing | Full methods and assumptions in the PDFs

EXPERIMENT DESIGN SIMULATOR

Change any input | Sample sizes update immediately | Two-sided alpha = 5%

BASELINE CONVERSION

Value

1.8%

TARGET RELATIVE LIFT

Value

20%

STATISTICAL POWER

Value

80%

ALLOCATION TO ADS

Value

50%

TOTAL USERS REQUIRED

47,014

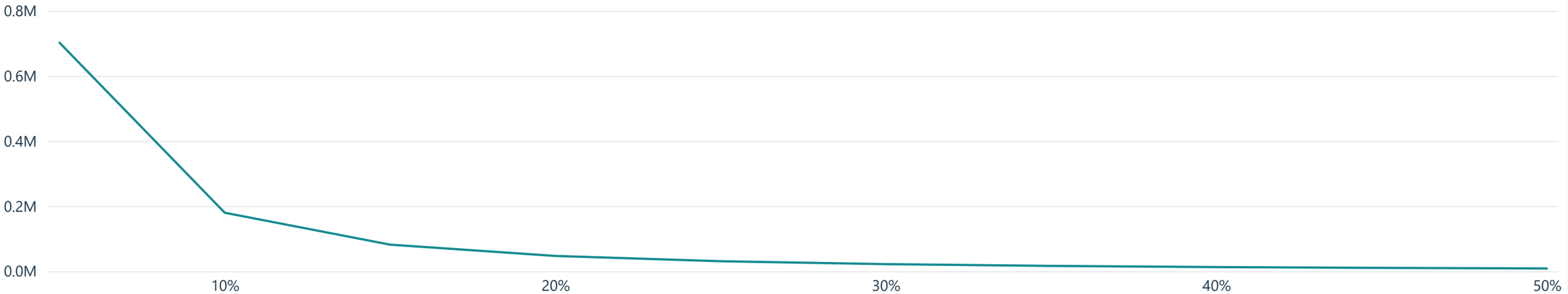
PSA USERS

23,507

AD USERS

23,507

SENSITIVITY | REQUIRED USERS BY RELATIVE LIFT



Normal approximation; independent users, fixed horizon, no attrition. Curve varies lift; other inputs follow your selections.

Choose an input from each dropdown. Defaults: 1.8% baseline / 20% relative lift / 80% power / 50% ads.

EVIDENCE

Full cohort | Design validity still requires assignment and measurement logs

PRIMARY P-VALUE

1.71E-13

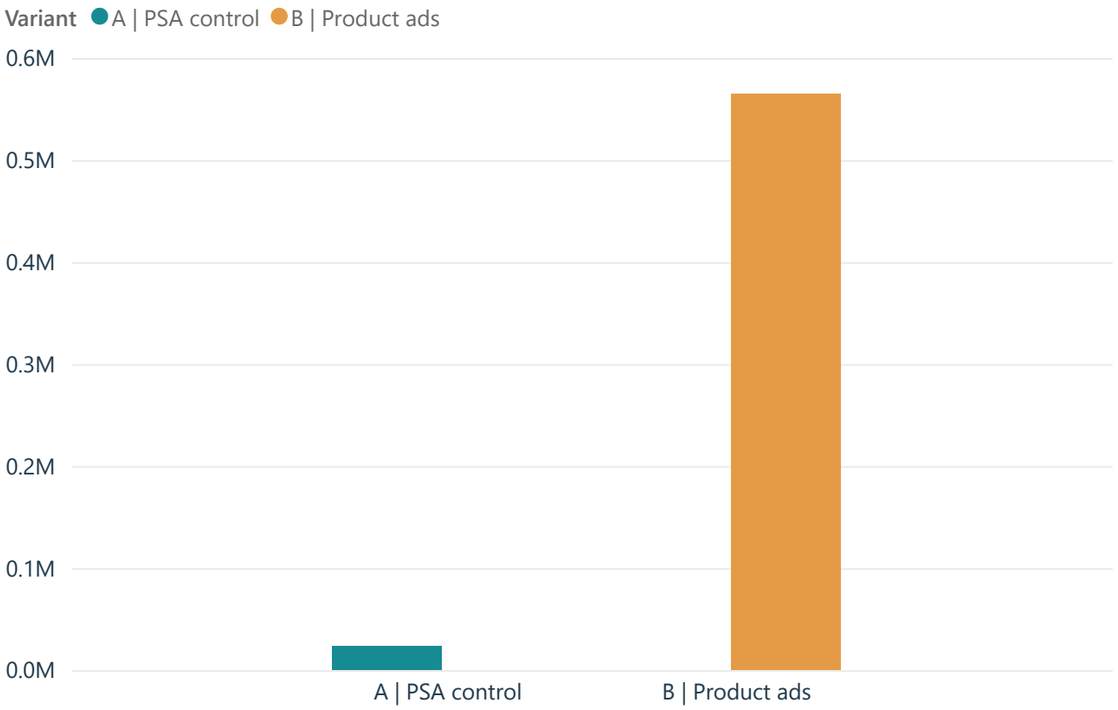
RISK RATIO

1.431

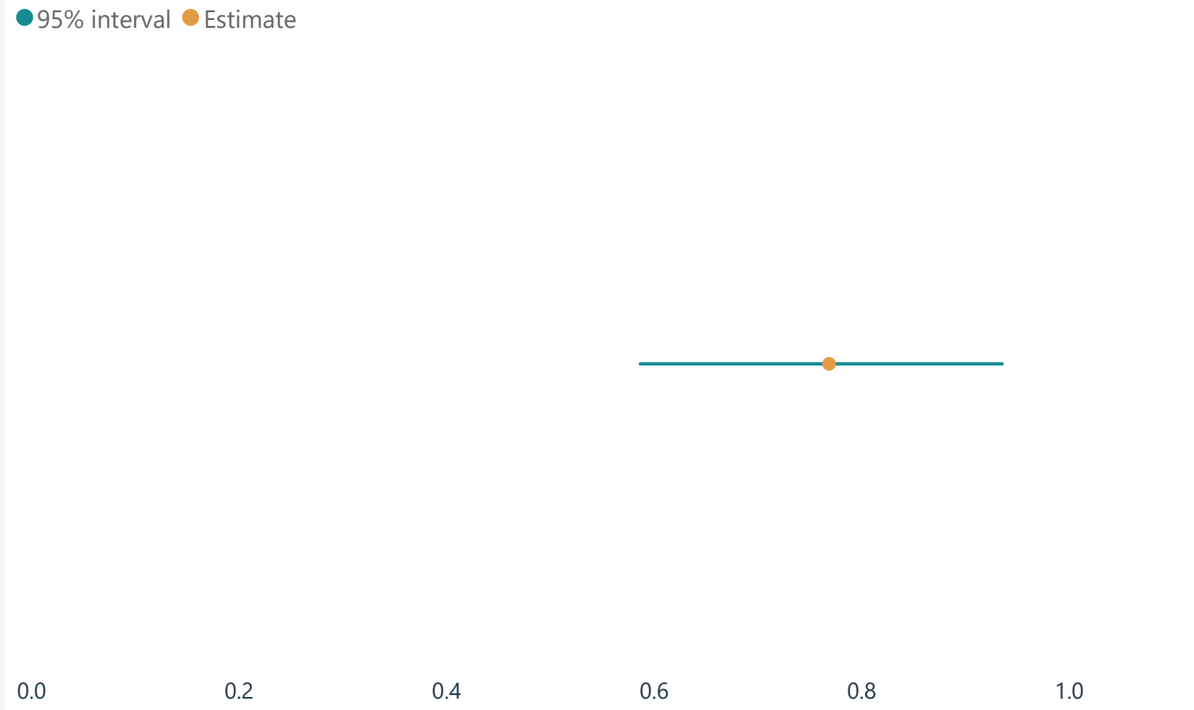
VARIANCE FROM PSA ARM

94%

OBSERVED ALLOCATION | USERS



95% CONFIDENCE INTERVAL | ABSOLUTE LIFT (pp)



Unequal allocation is not a failure without the planned split. Details and sensitivity checks: analysis report.

Kaggle: faviovaz/marketing-ab-testing | Full methods and assumptions in the PDFs

EXPOSURE EXPLORER

Exposure was not randomized | Explore patterns, not a causal dose response

PEAK WEEKDAY

Peak day

All

OBSERVATIONAL ASSOCIATION / More exposure does not prove more conversions

USERS IN VIEW

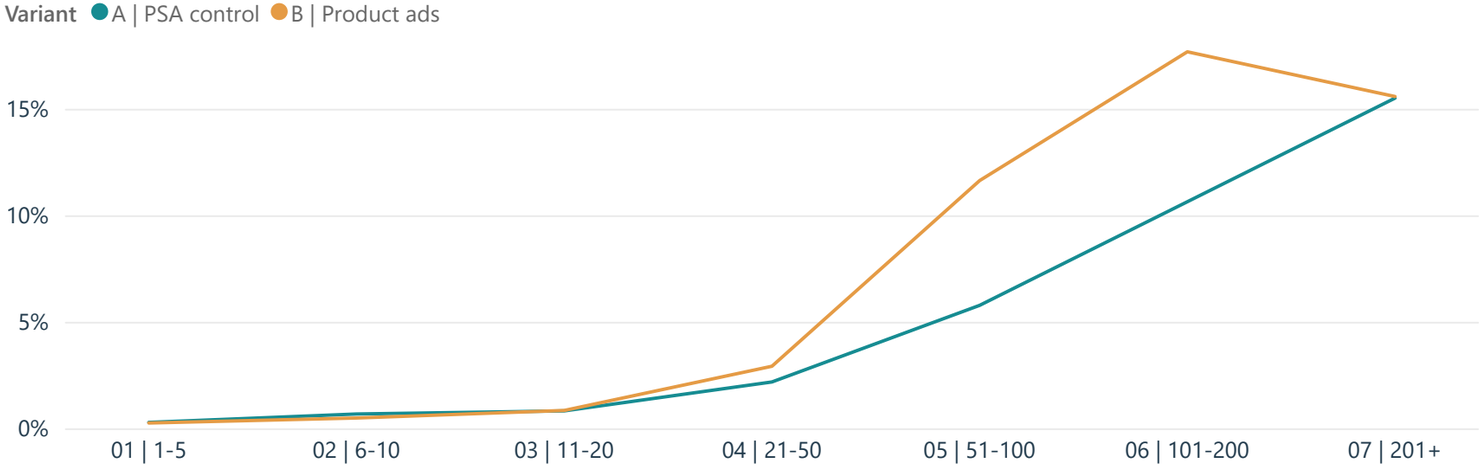
5,88,101

IMPRESSIONS

1,45,97,182

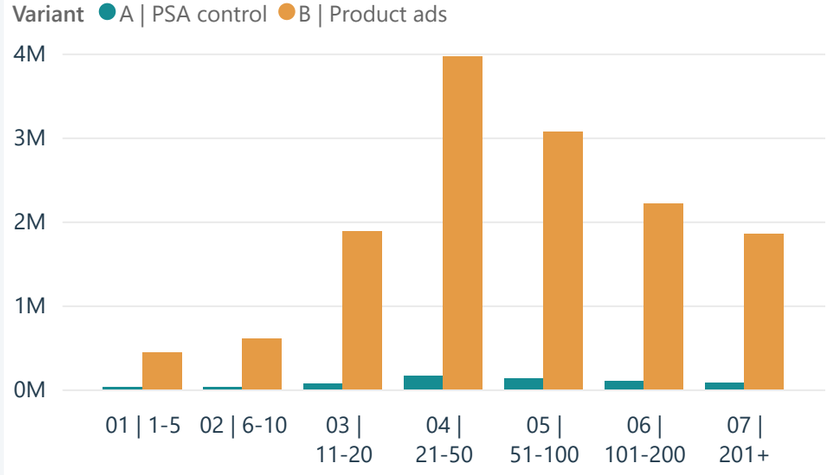
CONVERSION BY OBSERVED FREQUENCY

Variant ● A | PSA control ● B | Product ads



IMPRESSION CONCENTRATION

Variant ● A | PSA control ● B | Product ads



TOP ~1% OF USERS / 13% of recorded impressions

Engagement and targeting can affect both exposure and conversion. Test randomized frequency caps before changing delivery.

Kaggle: faviovaz/marketing-ab-testing | Full methods and assumptions in the PDFs

TIMING EXPLORER

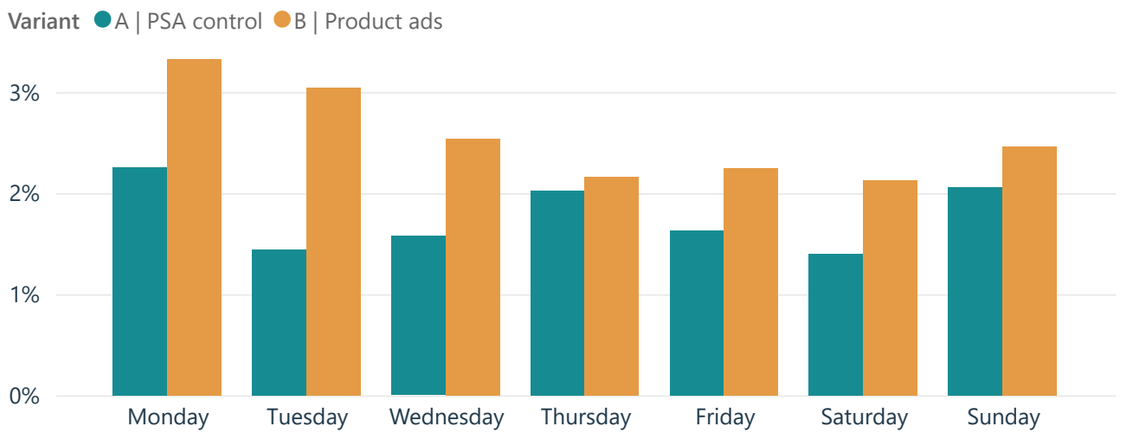
Peak-exposure context | Select a weekday to compare hourly profiles

PEAK WEEKDAY

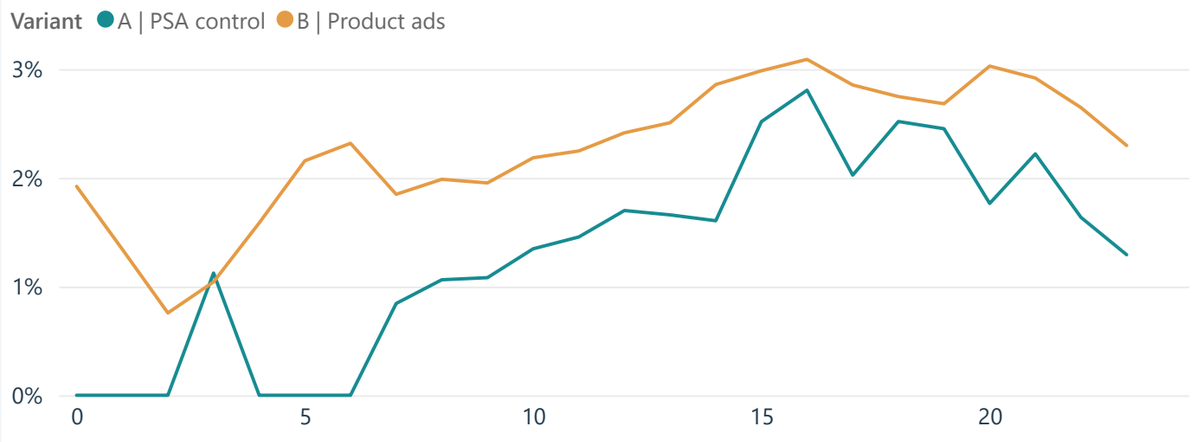
Peak day

All

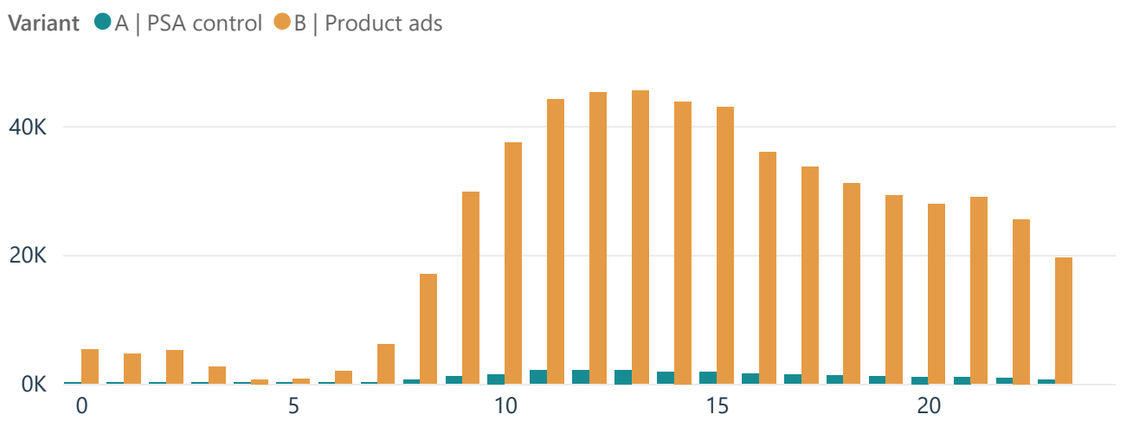
CONVERSION BY PEAK WEEKDAY



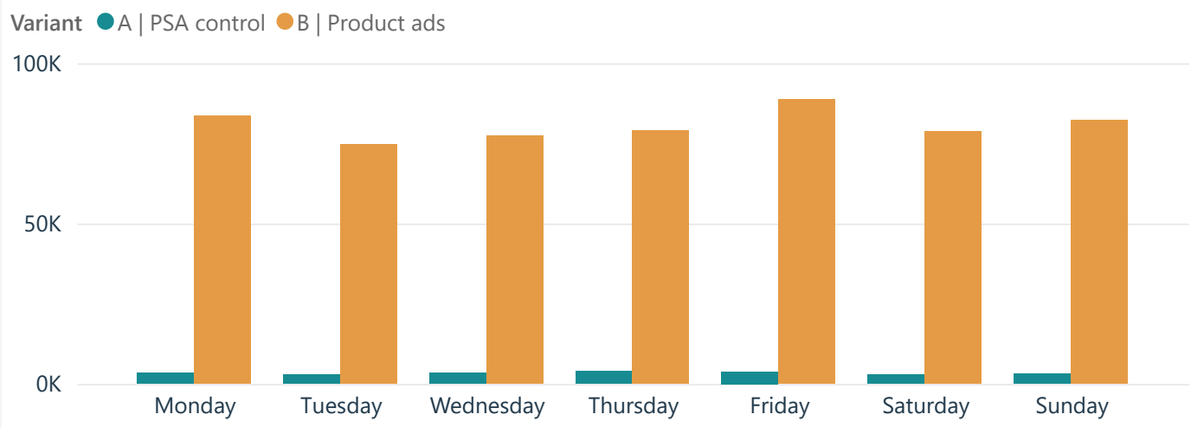
CONVERSION BY PEAK HOUR



HOURLY DENOMINATORS



WEEKDAY DENOMINATORS



Peak exposure time, not conversion time. Descriptive patterns; no causal scheduling recommendation.

Kaggle: faviovaz/marketing-ab-testing | Full methods and assumptions in the PDFs